



16-18, Knowledge Park II, Greater Noida, Uttar Pradesh 201306

Commonwealth Bank

Summer Internship Project Presentation

Introduction to Software Engineering Job Simulation

**Building a Financial Cybersecurity Web Portal & Designing Secure Cloud
Architecture**

Student Information

Name

Shubham Chaudhary

Enrollment Number

25scs1003004870

Program & Semester

B.Tech 3rd Semester (2CSE27)

Institution

IILM University, Greater Noida



Internship Overview & Objectives



Real-World Simulation

Stepped into the role of an entry-level software engineer at Commonwealth Bank via the Forage Virtual Internship program

Web Development Foundations

Built an educational web portal titled "Tips for Financial Cybersecurity" using raw HTML5 and CSS3

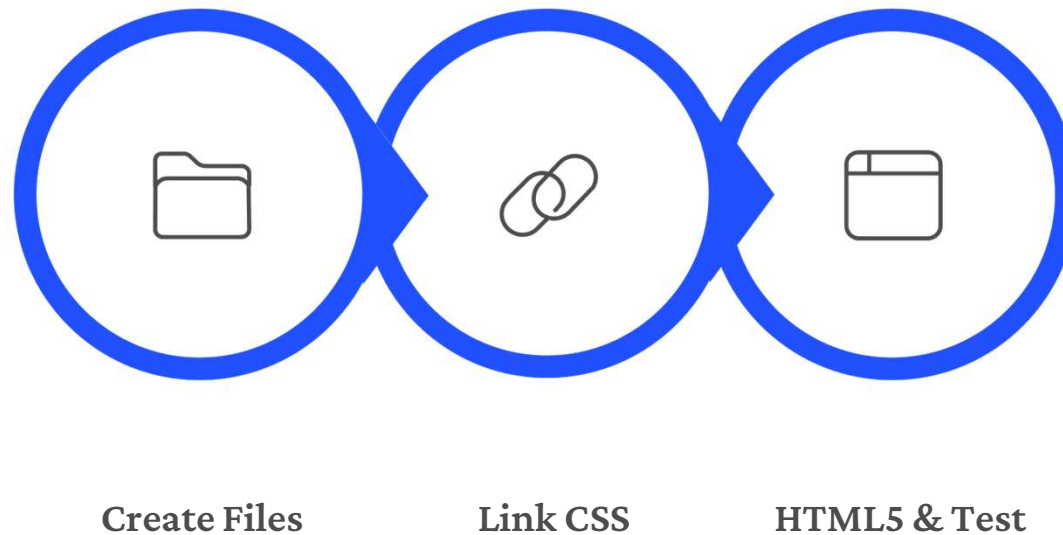
Brand Guideline Compliance

Customized web pages to match Commonwealth Bank's official typography, color scheme, and visual layout standards

Cloud & Hosting Proposal

Formulated an enterprise-grade web hosting and security strategy to support future client portal expansions

Task 1: Environment & Project Setup



A clean, modular project architecture was established from the outset — separating structure and styling concerns to ensure scalable, maintainable code aligned with professional engineering standards.

Task 2: Financial Cybersecurity Content



Strong Passphrases

Multi-word phrases with special characters; strictly no credential reuse across accounts



Multi-Factor Authentication

App-based 2FA to prevent unauthorized access even if passwords are compromised



Phishing Prevention

Banks never ask for PINs or full credentials via unsolicited messages — client education is key



Public Network Safety

Avoid financial transactions on public Wi-Fi without VPN encryption

Content strategy: condensed financial security research into a scannable 350-word guide for retail banking clients, utilizing structured `<div>` and `<section>` tags.

Task 3: Commonwealth Bank Brand Guidelines

Color Palette

Typography Hierarchy

- **Font Family:** Gill Sans / Lucida Sans (Single Sans-Serif)
- **Title:** 70pt — **Subheadings:** 38pt — **Body:** 18pt
- **Text Color:** Standard #1E1E1E; pure white (#FFFFFF) on black containers only

Implementation Standards

Card Component Architecture

CSS Flexbox card layouts with left accent borders matching brand colors

Visual Hierarchy

Distinct separation between header banner, hero statement, and card tip containers

Contrast Compliance

Fully compliant with WCAG accessibility contrast ratios across light and dark elements

Primary Yellow

#FFCC00

Brand Black

#1E1E1E



Task 4: Secure Web Hosting Proposal



TLS 1.3 / HTTPS

End-to-end transport encryption with HSTS headers to prevent eavesdropping and Man-in-the-Middle (MitM) attacks



WAF & DDoS Defense

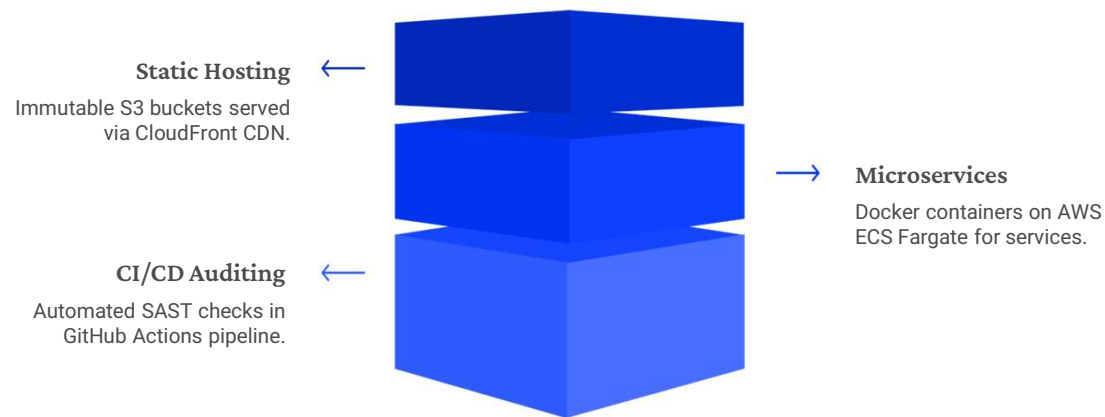
Cloudflare / AWS CloudFront WAF to mitigate Layer 3/4 volumetric attacks and filter Layer 7 XSS and SQL injection attempts



IAM & Access Control

Role-Based Access Control (RBAC), enforced hardware MFA, and SSH key whitelisting for deployment administrative pipelines

Infrastructure & Cloud Architecture



Architecture Layers

1 Static Hosting Phase

Immutable AWS S3 bucket storage
fronted by CloudFront CDN distribution
for global low-latency delivery

2 Containerized Microservices

Docker container deployments via AWS
ECS Fargate for future scalable API
endpoints

3 CI/CD Automated Auditing

Static code analysis (SAST) integrated
directly into GitHub Actions workflows
for continuous security assurance

Risk Mitigation & Compliance Matrix

Security Challenge	Associated Threat	Strategic Mitigation
Dynamic Banking Expansion	PII & financial data leakage	AES-256 database encryption at rest & Zero-Trust API gateways
Ransomware / Server Outage	Data loss & service downtime	Multi-region automated immutable backups & DR runbooks
Regulatory Non-Compliance	Breaching APRA CPS 234 standards	Centralized audit logging (AWS CloudTrail) & automated SIEM alerts

Key Skills & Learning Outcomes



1

Technical Development

Mastered semantic HTML structure, modular CSS architecture, responsive flex layouts, and strict brand style adherence

2

Technical Communication

Developed skills to translate complex security concepts into simple, actionable guidance for retail banking clients

3

Cybersecurity & Engineering

Gained understanding of secure hosting frameworks, regulatory compliance (APRA CPS 234), and enterprise cloud controls

GitHub Repository Submission



Repository Name (Public):

2025-26_Shubham_Chaudhary_25scs1003004870_3rd_Semester_2CSE27

Submitted Documents



Internship Report

Internship_Report.pdf

✓ Ready



Internship Presentation

Internship_Presentation.pdf

✓ Ready



Completion Certificate

Internship_Completion_Certificate.pdf

✓ Ready

